Emotional Reactivity in Adults with Childhood Exposure to 9/11 Trauma.

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Introduction

- Childhood exposure to trauma has enduring detrimental effects on mental health.
- Adults who experienced the 9/11 WTC terror attack during childhood exhibit higher rates of psychiatric disorders, notably anxiety and panic.
- Anxiety disorders are characterized by a tendency to misinterpret harmless stimuli as threats, and difficulties regulating emotions, leading to heighten emotional reactivity.
- Neuroimaging studies indicate that this emotional dysregulation stems from impairment in attention, emotion regulation and cognitive control systems.

Aim

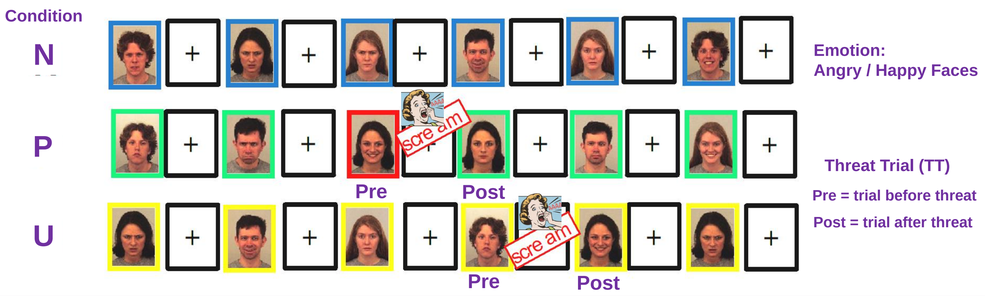
Examine behavioral and brain activation differences in threat reactivity between exposed groups with and without mental health disorders.

We expect that individuals exposed to the 9/11 terror attack during childhood will exhibit heightened emotional reactivity to threat, particularly those with psychiatric disorders.

Behavioral Task and Assessments

NPU task: we used a modified version of NPU task with screams as aversive (500 ms) stimuli.

- Subjects occasionally hear the scream while they categorized faces as angry or happy.
- A color frame during face presentation indicated the condition: no threat (Neutral), threat after current face (Predictable) or threat after any face (Unpredictable).
- Faces (500 ms) were followed by a fixation cross (2-6 s).
- Conditions were presented in interleaved blocks, separated by a fixation cross (10-12 s) and instruction screen (5 s)



Current Mental Health Assessments: PHQ-8, GAD-7, PCL, PDSR, STAI and IUS.

Sample

49 adults (26 F; 26-39 y/o, mean age at 9/11/2001= 11.5+- 3.8 y/o) with or without 9/11 exposure and past mental health disorders:

- exposed with mental health disorders (N=20)
- exposed without mental health disorders (N= 18)
- not exposed without mental health disorders (N=11)

MRI Acquisition and Preprocessing

2 EPI runs (TR: 800 ms , TE: 30 ms, 52° flip, FOV: 216 mm, 2.4×2.4×2.4 mm, 60 axial slices, 635 volumes, 8 min), and 2 high T1 MPR images (TR: 2400 ms, TE: 2.15 ms, 8° flip, FOV: 256 mm,1.0×1.0×1.0 mm) were acquired with multiband on 3.0 T Siemens scanner with a 32-channell head coil. Standard pre-processing was done in SPM12.

Analyses

Behavior. Performance and RT were compared between Groups, Conditions and Emotion for all trials and for Pre-Post Threat Trials using repeated mixed design ANOVA. Mental Health was compared between groups with single way ANOVA.

MRI. GLM analysis using SPM12. Single subject GLM included regressors for each condition, motion and signal artifacts (ArtRepair) and contrasts for U>N, P>N, and U+P+N. 8 subjects excluded (missing run or excessive artifacts). Whole brain BOLD signal compared between groups and conditions with ANOVA with multiple comparison correction (FWE p=0.05).

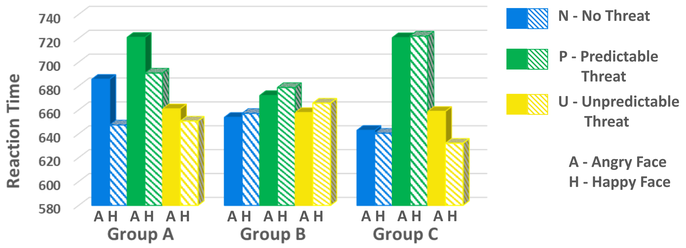
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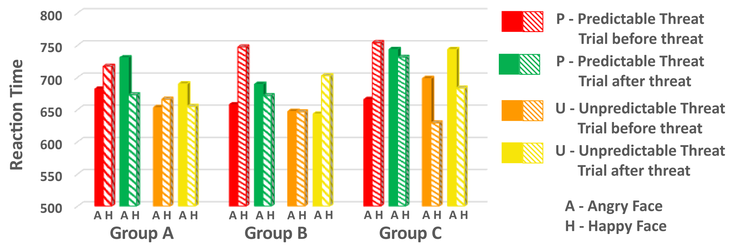
Thanks to all the participants for their time, to the GPEG team for their support with recruitment, to Charlie Brown III for technical support, and to Dr. Zhishun Wang for MRI scripts development. Thanks for support from NIOSH/CDC grant U01OH011694.

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Results

Behavioral Results





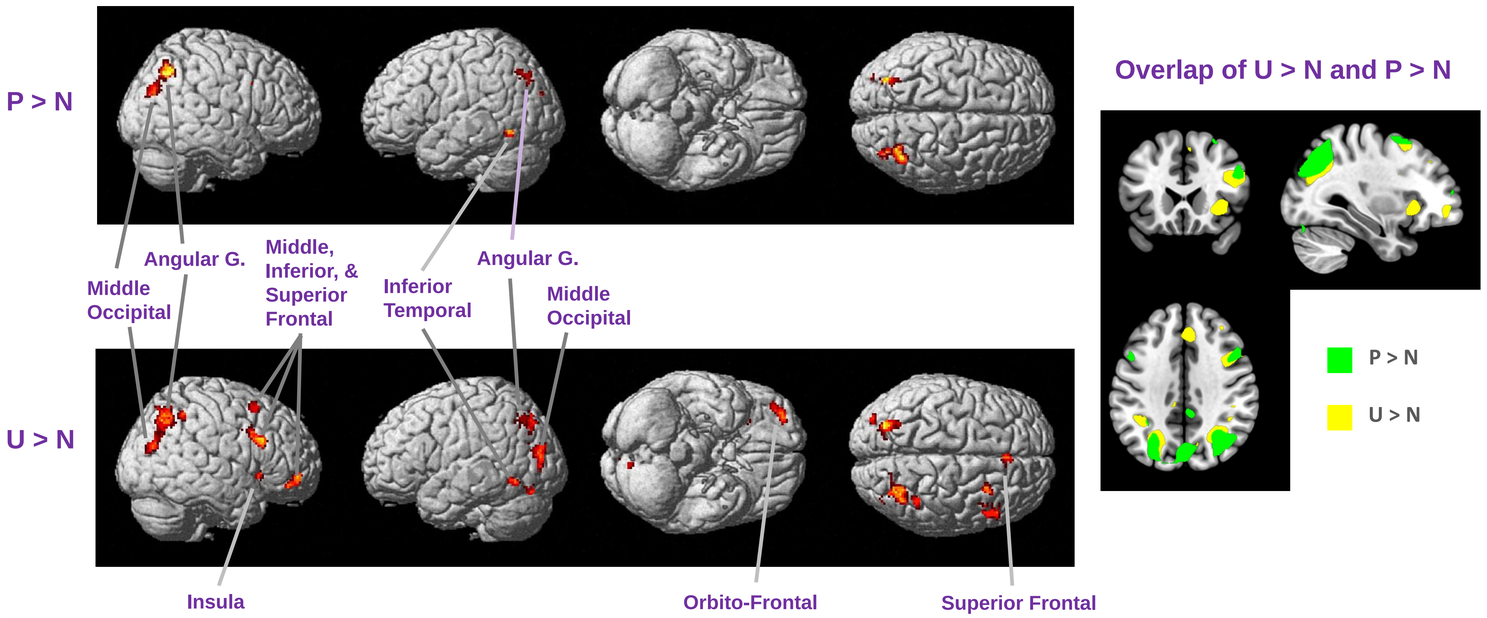
- Performance accuracy (~94%) in the NPU was similar across all Groups, Conditions and Emotions. Performance for Pre-Post Threat Trials was not affected by threat presentation.
- RTs differed significantly between Conditions [$F_{2,45}$=8.14, p<0.001], showing that RTs for P condition were larger than for N (p<0.003) and U (p<0.010) conditions.
- RTs also showed a significant Group x Emotion Interaction [$F_{2,46}$=3.61, p<0.035] indicating that RTs for angry faces were longer for group A.

- Significant Interaction of Condition x Emotion x Threat Trial [$F_{1,38}$=4.64, p<0.036] indicates that RTs differed for Pre-Post Threat Trials in P and U conditions. Overall, RTs were longer for P than U, and for trials Post-threat compared to those Pre-threat. For U condition RTs were longer Post-threat regardless of face Emotion. The largest effect was in the P condition, where RT was shorter for Pre-threat trials when the cue was presented with angry face, and shorter for Post-threat trials when the threat was followed by a happy face.

- Groups did not differ in demographics.

- Mental Health Assessments did not differ across Groups, but PTSD symptom count, which was subthreshold, was significantly greater for A.

Imaging Results



- No significant differences were observed between Groups.
- Brain activity differed significantly between Conditions, revealing increased engagement of parietal; frontal and occipital regions during U or P compared to N condition.
- U and P conditions show similar pattern of activation (Figure 4). However, activity in the Insula and Orbitofrontal gyrus was unique for U > N.

Conclusions

- High performance accuracy indicated ease in discriminating between negative and positive facial expressions.
- The slower RT in the P condition confirm the expected threat effect. We observed the faster RTs in trials with congruent threat cue and angry faces, while incongruent threat and happy faces caused interference leading to longer RTs. Additionally, angry faces post-threat elicited longer RT. These findings suggest a greater threat monitoring in the P condition.
- The engagement of the insula and orbitofrontal cortex, along with longer RTs for post-threat stimuli, suggest a greater aversive response in U compared to P. This was predicted for a sustained threat response even though RTs for U did not differ than those for N. All groups exhibited heightened activity in attention and control regions in response to the threat stimuli, indicating a similar processing for threat regardless of trauma exposure.
- Minimal differentiation in mental health symptoms among groups aligns with lack of significant differences between groups in both behavioral outcomes and brain signals in the NPU task.